

TOMATO DISEASE CLASSIFIER – PROJECT REPORT

Project Summary

This project develops a CNN classifier for tomato disease detection using the PlantVillage dataset, classifying images as Healthy Tomato or Tomato Yellow Leaf Curl Virus. The model uses MobileNetV2 transfer learning trained for 5 epochs, achieving 100% test accuracy on 98 test images.

The Streamlit application allows users to upload tomato leaf images (JPG, PNG, JPEG) and receive instant predictions with confidence scores. Key challenges included TensorFlow installation incompatibilities on Streamlit Cloud and Keras version conflicts. These were resolved through careful dependency management and model format optimization.

Future improvements include deploying to alternative cloud platforms and integrating real-time camera feed support for field-based diagnosis. The model demonstrates high reliability for practical agricultural applications.

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